NitroMan - Speed Optimization Engine

Version: 1.0.0 **Author:** Ajaypal Singh

NitroMan is a lightweight yet extremely powerful WordPress performance optimization plugin designed to make websites blazing fast without breaking functionality. It implements multiple layers of optimization, from HTML/CSS/JS minification to caching, database cleanup, and fine-tuned WordPress tweaks to provide a comprehensive Core Web Vitals boost.

🚀 Core Features & Capabilities

1. Advanced Page Caching

NitroMan intercepts page requests and generates static HTML files, bypassing costly PHP processing and database queries.

- **Lightning-Fast Delivery:** Serves pre-built HTML from a dedicated cache directory (`/wp-content/cache/nitroman/`).
- **GZIP Compression Compatibility:** Ready for server-level compression to minimize payload size.
- **Smart Cache Busting:** Automatically purges the cache when updating posts, changing themes, or modifying global settings.
- **Preloading Engine:** Automatically preloads pages to ensure the cache is always warm and ready for visitors.

2. Deep Image Optimization

Images are often the largest payload. NitroMan handles images natively to prevent layout shifts and save bandwidth.

- **Native Lazy Loading:** Defers loading of off-screen images until the user scrolls near them.
- **Automatic Dimension Enforcement:** Injects missing `width` and `height` attributes to prevent Cumulative Layout Shift (CLS), a key metric for Google rankings.
- **WebP Detection:** Ready to detect and serve next-gen image formats when available.

3. Comprehensive Asset & Script Management

NitroMan ensures JavaScript and CSS don't block the initial render.

- **Intelligent Defer/Async:** Automatically adds `defer` or `async` tags to non-critical scripts.
- **Safety First:** Built-in exclusions are hard-coded for core scripts (e.g., jQuery) to ensure themes and complex plugins do not break.
- **Query String Removal:** Removes version strings (`?ver=x.x`) from static resources to improve caching efficiency on CDNs and edge servers.

4. HTML & CSS Minification

Reduces the sheer volume of code sent over the wire.

- **HTML Minification:** Strips unnecessary whitespace, comments, and line breaks from the rendered HTML document.
- **Critical CSS Management:** Prepares the layout for inline vital CSS, deferring the rest, significantly boosting the First Contentful Paint (FCP).

5. Automated Database Cleanup

A bloated database slows down queries. NitroMan efficiently sanitizes the WordPress tables.

- **Revision Control:** Deletes unnecessary post revisions and auto-drafts.
- **Spam & Trash Removal:** Automatically permanently deletes spam comments and items in the trash.
- **Transient & Meta Fixes:** Cleans up expired transients and orphaned post metadata to maintain peak database performance.

6. Core WordPress Tweaks

Disables legacy features and redundant WordPress behaviors.

- **Disable Emojis:** Prevents WordPress from loading extra emoji scripts/styles if they are not needed.
- **Disable OEmbeds:** Stops the site from generating external embed codes and loading unneeded JavaScript.
- **XML-RPC Protection:** Option to disable XML-RPC to secure against DDoS attacks.
- **Heartbeat Control:** Throttles or disables the WordPress Heartbeat API in the frontend and admin to save server CPU cycles.
- **Head Cleanup:** Removes generator meta tags (e.g., `<meta name="generator" content="WordPress x.x">`) for better security and a lighter Document Object Model.

7. Pre-Connection & Resource Hints

Informs the browser to establish early connections to important external services.

- **DNS Prefetching:** Instructs the browser to perform DNS lookups in the background for predefined URLs (like `fonts.googleapis.com` or CDNs).
- **Preconnect:** Establishes early HTTP connections to critical external domains to eliminate latency.
- **Font Preloading:** Safely preloads selected font files (`.woff2` , `.ttf`) so they are ready the instant they are required by the CSS.

🛡️ Built for Peak Stability and Security

Recursion Guards & Memory Safety

NitroMan uses advanced OOP design patterns and state flags (`is_processing`) across all its content-filtering modules (like the HTML Minifier) to ensure that large pages never trigger memory exhaustion or infinite loops during the optimization phase.

Security Implementation

- **Hardened AJAX/Admin actions:** Strict nonce verification utilizing `Security::verify_request()` .
- **Capability Boundaries:** All administrative functions enforce the `manage_options` capability.
- **Secure File Operations:** Dedicated index files guard the cache directories against unauthorized directory traversal or execution attacks.

💻 The Administrator Experience

The plugin features a professionally crafted, modern UI within the WordPress administrative backend (`/admin/views/dashboard.php`).

- **One-Click Toggles:** Easily enable/disable specific optimization engines.
- **Clean Settings Panels:** Grouped intelligently with clear descriptions.
- **Real-time Status:** Dashboard visualizes cache status, database health, and active optimizations.